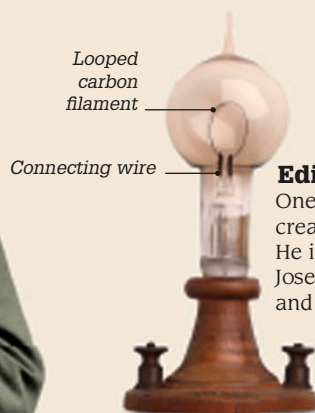
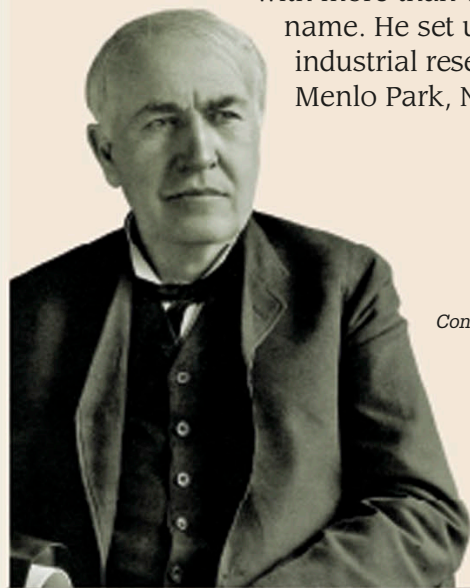


1847–1931 THOMAS EDISON

American scientist Thomas Edison began work as a telegraph operator at the age of 15. He went on to become one of the most famous inventors of all time, with more than 1,000 patents to his name. He set up the world's first industrial research laboratory at Menlo Park, New Jersey.



Edison's light bulb

One of Edison's best-known creations is the electric light bulb. He improved on English chemist Joseph Swan's design (see p.139), and produced a pure vacuum inside the glass bulb, while using a longer-burning, carbonized bamboo fiber for the filament.



Edison's home phonograph, a later, mass-produced home player, c 1898

First sound recording

Edison invented the phonograph in 1877. A needle in the instrument engraved the vibrations of the human voice on a rotating tinfoil cylinder. To play the sound back, the needle retraced the grooves.

1885

1881

First electric tramway

The first electric tramway was opened in Lichterfelde, a suburb of Berlin, Germany. Each car was equipped with an electric motor, with the current supplied via the running rails. People and horses often received electric shocks at railroad crossings before an overhead wire was introduced in 1891.

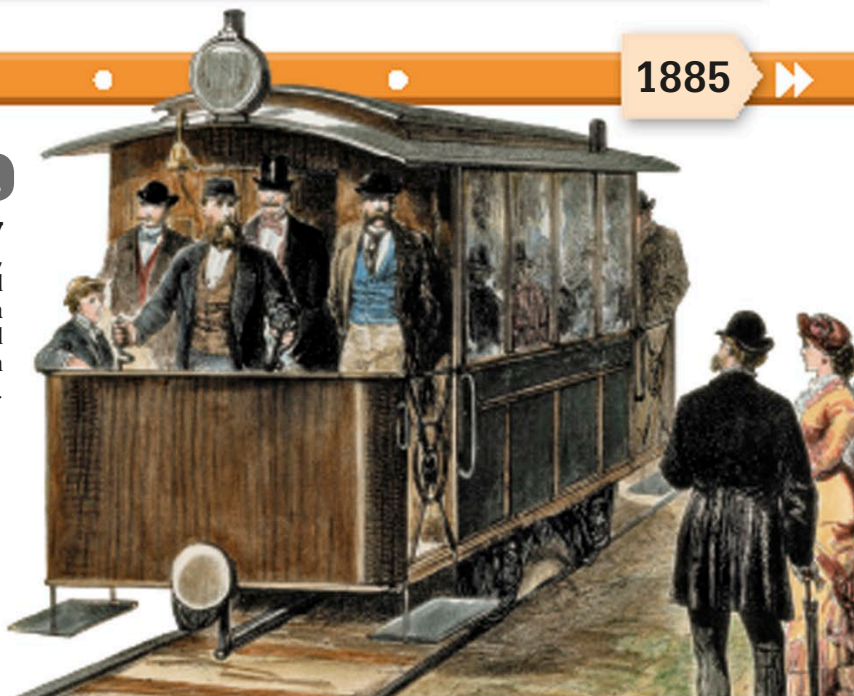


Illustration of the first electric tramway

1881

Brighter cities

Although arc lights (see p.114) had lit up a number of large cities since the 1870s, the small town of Godalming, UK, caught the attention of the world when it became the first place to install electric light bulbs. The electric supply, which also lit houses, was generated by a water-powered dynamo.

Moving horse

Photographs taken by English photographer Eadweard Muybridge revealed that a trotting horse has all four legs off the ground at the same time.



“We will make electric light so cheap that only the rich will be able to burn candles.”

Thomas Edison, 1879